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Predicting Acute Food Insecurity Risk in Kenyan Counties

Using climate and market signals as leading indicators of IPC Phase 3 (Crisis), an early-warning model for Kenya's National Drought Management Authority

Data Alchemists Patrick Njoroge · Chelimo Ivy · Alex Njihia · Joel Andrew · Halima Ahmed · Rebecca Monchari

BUSINESS UNDERSTANDING

Kenya's food security response is reactive

10M+

Kenyans live in arid & semi-arid lands (ASAL), dependent on two seasonal rains

5

consecutive failed rainy seasons (2020–23), the worst drought in 40 years

4.5M

Kenyans pushed into acute food insecurity at the drought's peak

4–8 wks

lag before official IPC classifications confirm a crisis on the ground

By the time a county is officially classified in Crisis, households have already skipped meals, sold productive assets, and exhausted their coping capacity.

THE PROBLEM WE ARE SOLVING

Crises are confirmed after the damage, yet two signals arrive weeks earlier

Today: detection, not prediction

- IPC Phase 3 (Crisis) is the official threshold at which NDMA declares a food emergency and activates aid.
- Classifications rely on field assessments and are published 4–8 weeks after conditions have deteriorated.
- Aid agencies get no lead time to pre-position resources, so response arrives after coping capacity is gone.

Two leading signals, currently unused in any predictive model

Rainfall failure: CHIRPS satellite anomalies every 10 days, weeks before crop and livestock losses are visible.

Food price spikes: WFP monthly market prices reveal collapsing purchasing power before hunger becomes measurable.

Prediction question: Can rainfall anomalies + food prices predict whether a Kenyan county reaches IPC Phase 3+ in a given month, before the official classification is published?

Who this serves, and why it matters

Kenya NDMA

County-level crisis probabilities weeks before IPC release, enabling early warnings and pre-positioned food aid.

WFP Kenya

Longer planning horizon for procurement, logistics coordination, and partner briefings.

Humanitarian NGOs

Target pre-crisis interventions to high-risk zones before deterioration passes the point of prevention.

Ministry of Agriculture

Prioritise drought-resistant seed and emergency livestock support to flagged counties ahead of a season.

Impact: a shift from reactive to predictive response: earlier mobilisation, precise county-level targeting, cheaper prevention over emergency response, and an architecture scalable across East Africa.

SUCCESS METRICS

Targets set before any modelling began

Metric	Target	Rationale
Recall (crisis class)	**≥ 0.75**	Primary metric, since a missed crisis costs far more than an unnecessary alert
Precision (crisis class)	**≥ 0.60**	Prevents alert fatigue and wasted field-verification trips
PR-AUC (crisis class)	**≥ 0.75**	Imbalance-robust, as crisis prevalence is only ~13–20%
F1 (macro)	**≥ 0.70**	Penalises models that ignore the minority crisis class
ROC-AUC	**≥ 0.80**	Overall discriminative ability across thresholds
Beat naive baseline	**PR-AUC > 0.197**	“Always no-crisis” scores 80% accuracy but catches zero crises

Recall first, precision second. A false negative, failing to flag a county heading into crisis, carries far greater humanitarian cost than a false alarm.

DATA

Three open datasets, merged to a county-month grain

TARGET

27,694 rows

FEWS NET IPC Classifications

Zone-level food insecurity severity, Phase 1 (Minimal) to 5 (Famine). Our target: Phase 3+ = crisis.

CLIMATE FEATURES

132,678 rows

CHIRPS Rainfall Anomalies

Dekadal (10-day) satellite rainfall incl. rfq, rainfall as % of long-term average, the key drought signal.

MARKET FEATURES

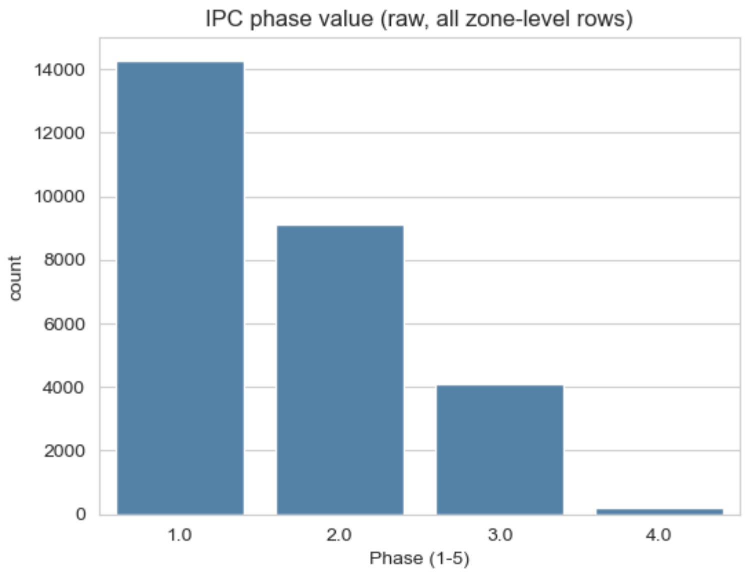
26,745 rows

WFP Food Prices, Kenya

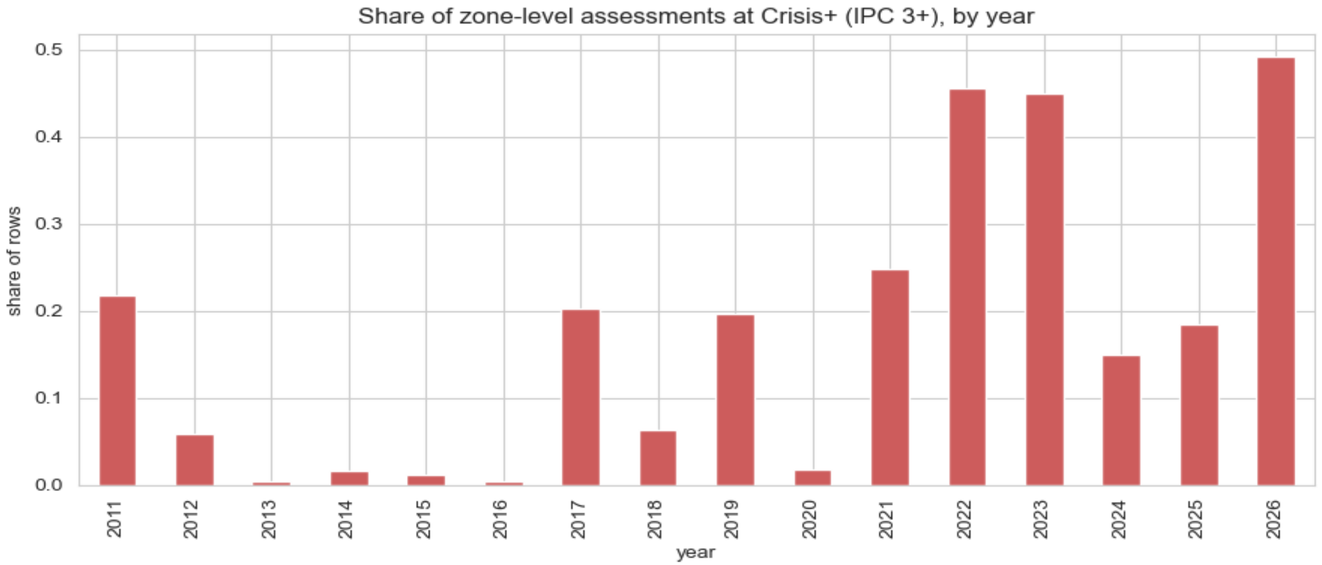
Monthly market prices for staple commodities (maize dominant) across Kenyan markets.

Preparation highlights: geographic harmonisation to counties · binary target construction (IPC 3+) · aggregation to county-month · lagged merge of features so predictors precede the outcome (leakage fix).

The target is rare, imbalanced, and tracks real droughts



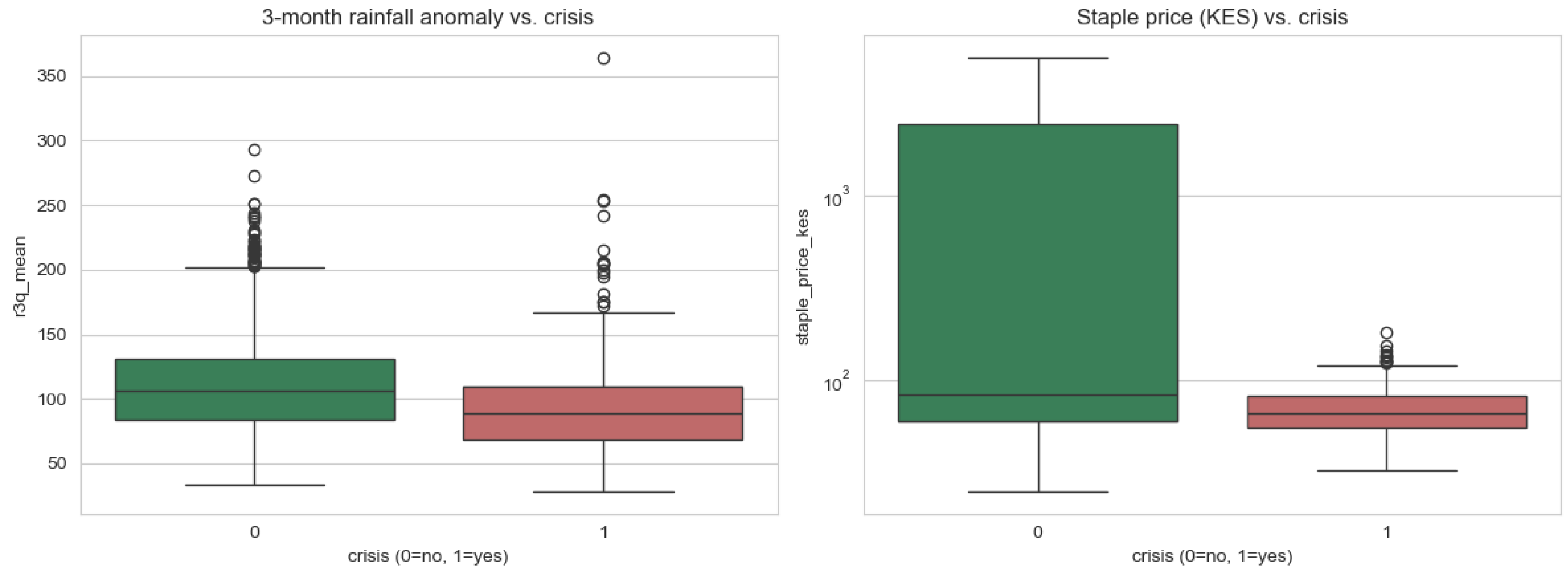
IPC phases skew heavily to Phase 1; Phase 3–4 are rare, Phase 5 absent



Share of Crisis+ assessments by year: spikes in 2011, 2017, 2021–22 droughts

Insights: crisis prevalence is only ~13–20%, so accuracy alone is misleading and recall must lead evaluation. Crisis rates surge exactly in Kenya's documented drought years, confirming the target is reliable and modellable.

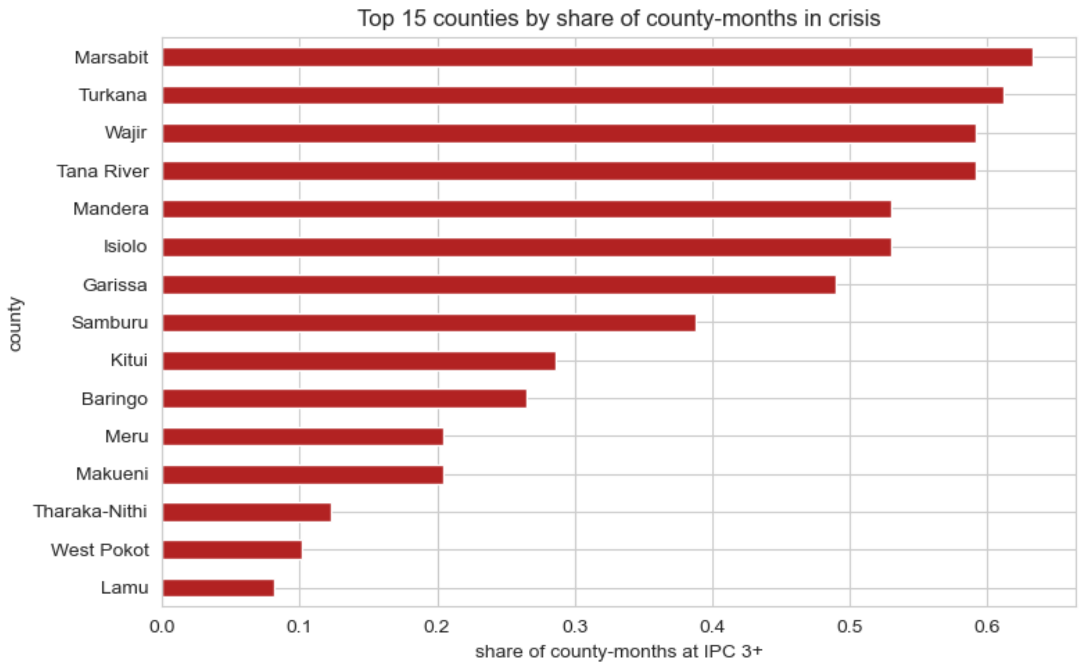
Rainfall is the strong signal; price level is the weak one



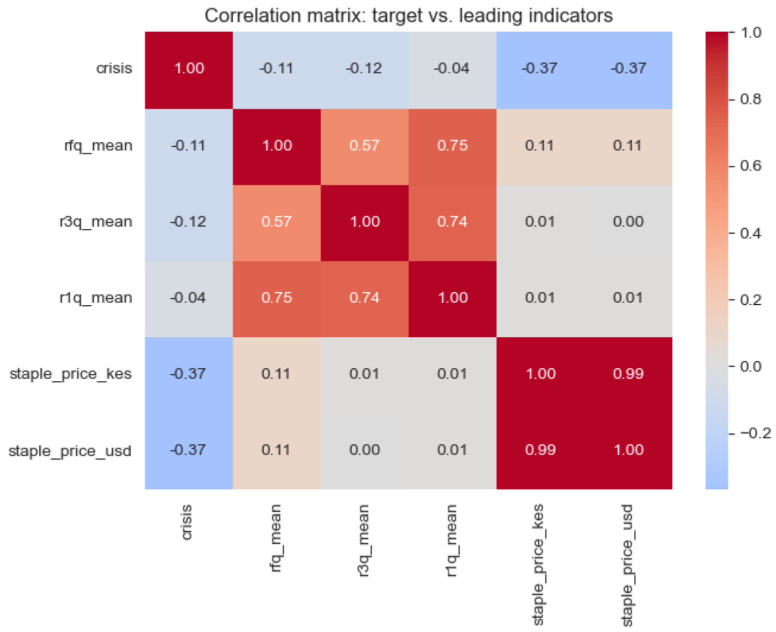
3-month rainfall anomaly (left) and staple price, log scale (right) vs crisis outcome

Insights: crisis months show clearly lower rainfall anomalies, and the 3-month window separates classes better than a single 10-day value because agricultural stress accumulates. Prices surprise us: absolute levels are not higher in crisis months, making price a supporting predictor only, with price trend the clearer path for future work.

Risk is geographically concentrated, and compound



Top 15 counties by share of county-months at IPC 3+: ASAL counties dominate



Correlation of target vs leading indicators: rainfall strongest (negative), prices weaker

Insights: Turkana, Mandera, Marsabit, Wajir, Garissa and Samburu lead, so location itself encodes structural vulnerability (justifying the is_asal feature). Crisis observations cluster where low rainfall and elevated prices coincide: a compound shock of reduced availability plus reduced affordability, so both signal families belong in the model.

Built for a public agency, not a tech lab

Chronological split, no leakage

Train on the past, validate, then test once on 2022+, with features lagged so predictors precede the outcome.

10 model families → 4 deployable finalists

Logistic Regression, Random Forest, XGBoost, LightGBM: all class-weighted, CPU-only, seconds to train, one small joblib file. SVM/KNN, deep nets and SMOTE variants dropped on deployment cost.

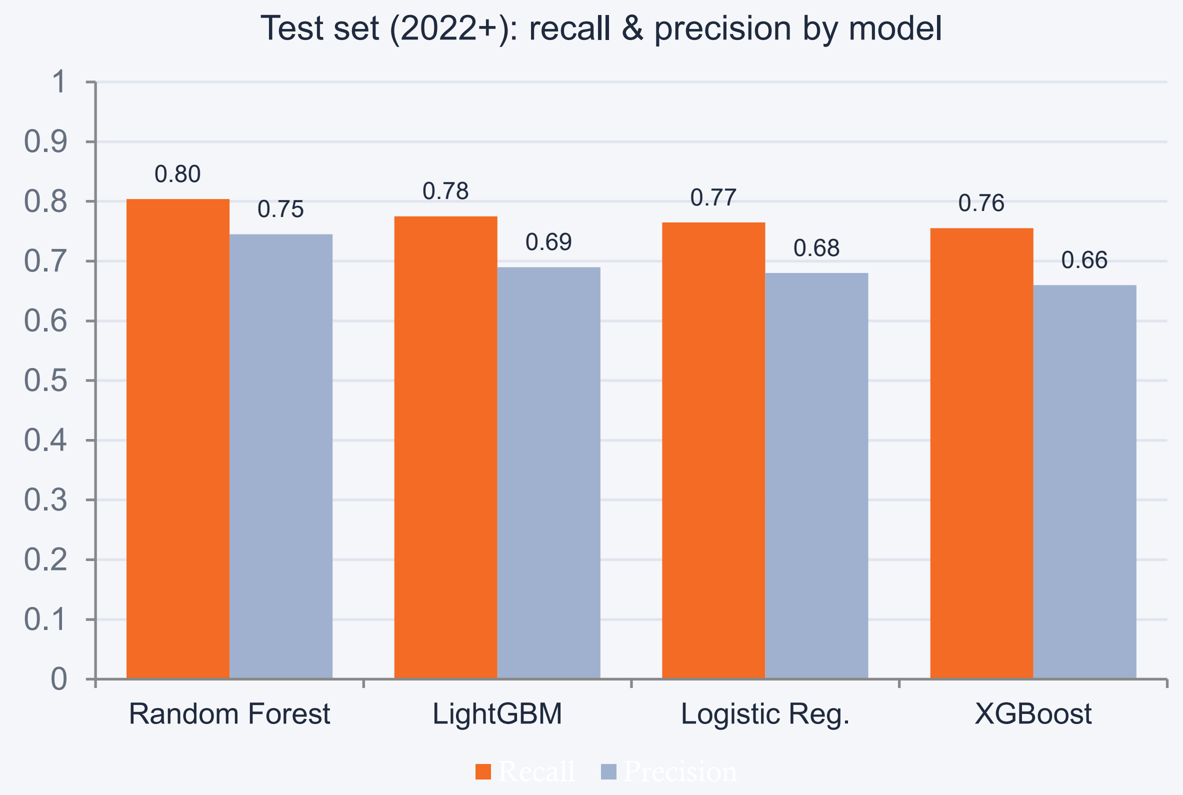
One fair threshold rule

On validation only: maximise precision subject to the Recall ≥ 0.75 humanitarian floor, giving the fewest false alarms among models that never miss the floor.

Why it matters: an analyst who stops trusting the alerts is a slower-acting analyst, and precision protects the recall floor's real-world value.

RESULTS

Random Forest wins, every pre-set target met on untouched test data

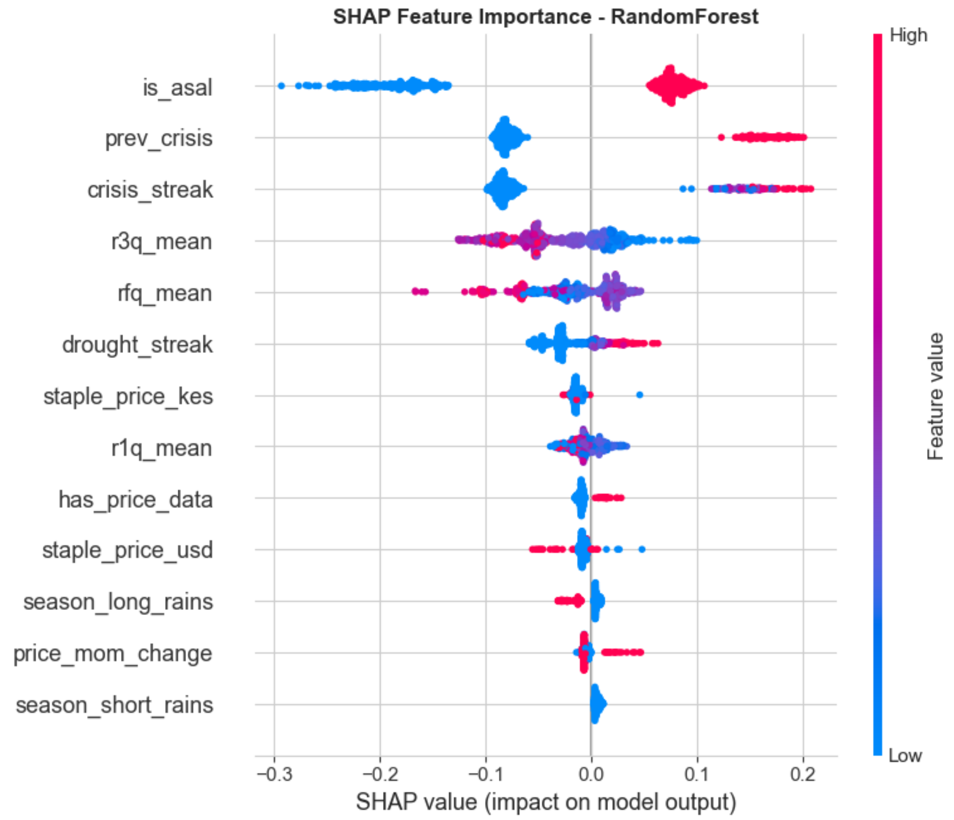


Random Forest, the deployed model

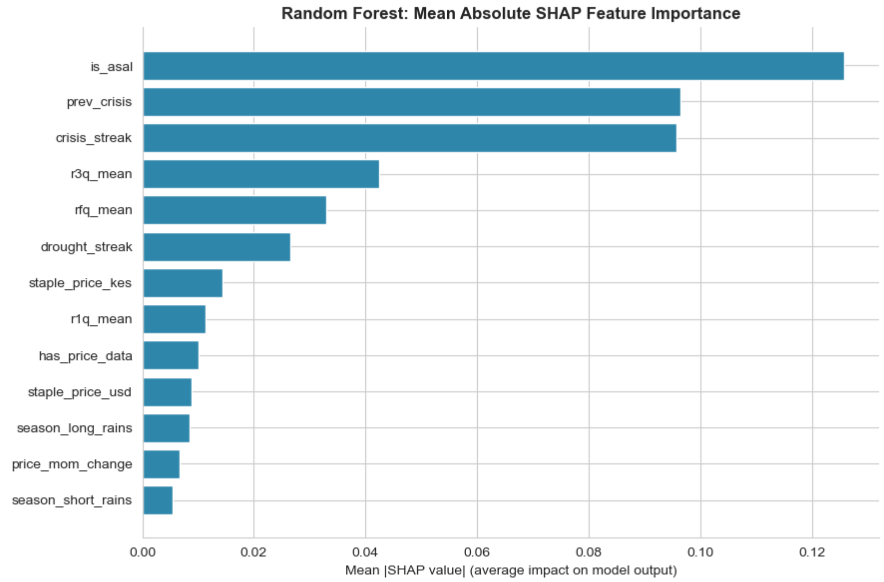
- All 6 success criteria met: Recall 0.80 · Precision 0.75 · PR-AUC 0.82 · F1-macro 0.86 · ROC-AUC 0.94, far above the 0.197 naive baseline.
- 82 of 102 crises caught with only 28 false alarms, roughly 3 crises caught per false alarm, an alert stream analysts can trust.
- **Live demo, October 2023: all 8 counties that reached Crisis+ were caught and ranked at the top of the bulletin.**

MODEL INTERPRETABILITY

What actually drives the predictions (SHAP)

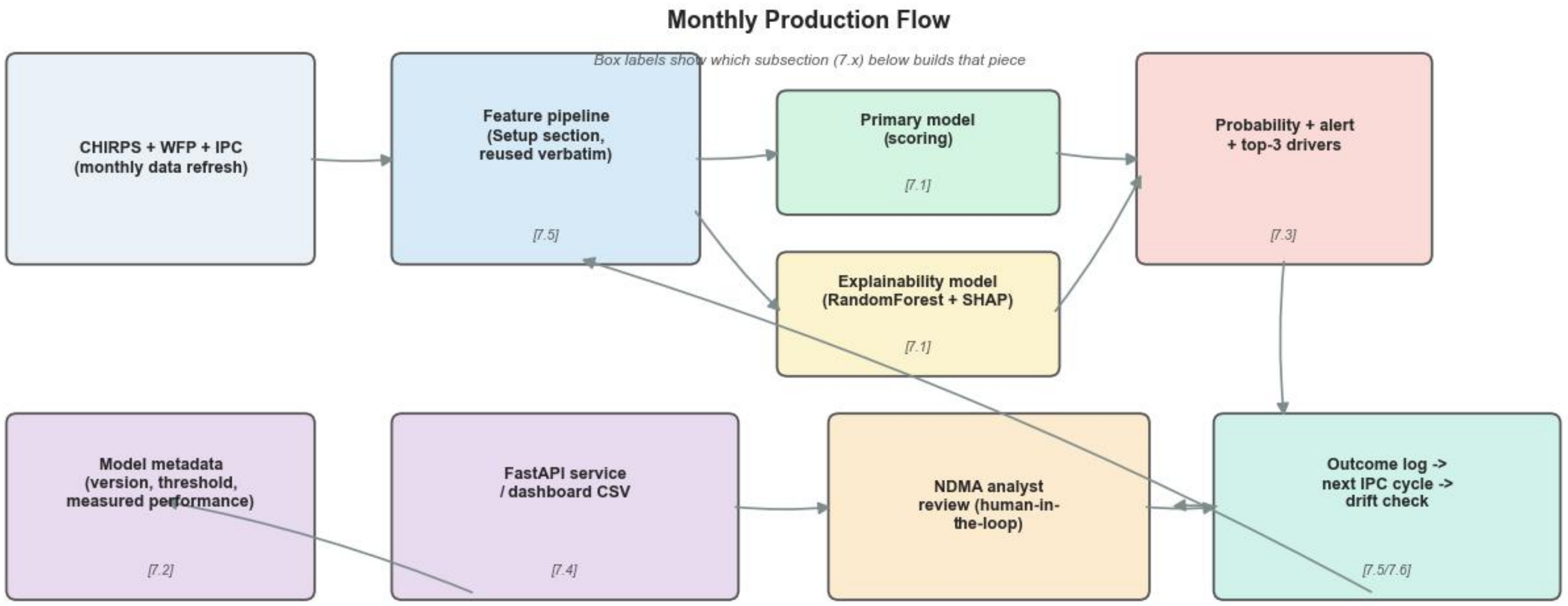


Per-prediction impact: red = high feature value, position = push toward / away from crisis



Reading the plots:
is_asal dominates, with crisis_streak and prev_crisis next: structural vulnerability and recent history lead. Rainfall (r3q, rfq) is the strongest condition-based signal; price level barely contributes, consistent with the EDA.

A monthly scoring pipeline with an analyst in the loop



Data flows left to right through scoring (top row), then down into serving, analyst review, and outcome logging

One run per month: new CHIRPS and WFP data refresh feeds the exact training-time feature pipeline; the model scores every county; SHAP attaches the top-3 drivers to each flag; the analyst reviews before any action is taken.

DEPLOYMENT INFRASTRUCTURE

Lightweight, versioned, and monitored

Versioned artifacts

Deployed model, fitted imputer and feature order saved as small joblib files, alongside a metadata JSON recording version, decision threshold and measured test performance.

Scoring service

A FastAPI service (or a plain dashboard CSV export) exposes score_new_month, the exact function a scheduled job calls; runs on any laptop CPU, no GPU or MLOps stack.

Human-in-the-loop dashboard

Each county lands on the NDMA dashboard as a crisis probability, an alert flag and a plain-language top-3 driver explanation. Analysts review; the model never triggers aid directly.

Monitoring & drift checks

Every prediction is logged and scored against the official IPC release 4–8 weeks later, catching performance drift before it becomes a missed crisis; monthly retraining slots in as new data lands.

Total footprint: a few MB of serialized files plus a scheduled monthly job, a realistic recurring cost for a public agency.

LIVE DEMONSTRATION

Scoring a real month: the October 2023 bulletin

County	Crisis probability	Actual outcome
Wajir	94.1%	✓ Crisis
Nyeri	93.8%	No crisis
Samburu	93.2%	✓ Crisis
Turkana	92.6%	✓ Crisis
Tana River	90.2%	✓ Crisis
Marsabit	90.1%	✓ Crisis
Mandera	89.8%	✓ Crisis
Garissa	89.4%	✓ Crisis
Tharaka-Nithi	87.2%	No crisis
Isiolo	85.8%	✓ Crisis

Top 10 of 18 flagged counties (of 47 scored), ranked by predicted probability.

8 / 8

counties that truly reached Crisis+ that month were caught, all ranked at the top of the bulletin

18 / 47

counties flagged for analyst review, a manageable verification workload

10

false alarms, concentrated in counties with genuinely elevated risk factors rather than spread randomly

Analysts open a ranked bulletin, not a model file.

INSIGHTS

What the model tells NDMA

Structural vulnerability dominates

is_asal is the single strongest driver: being an arid/semi-arid county alone pushes strongly toward crisis.

History matters

crisis_streak and prev_crisis rank next: recent crisis history carries independent predictive signal.

Rainfall is the top condition signal

Low r3q / rfq push toward crisis, as hypothesised, and the strongest of the "what's changing now" features.

Price level barely contributes

Consistent with EDA: price trend, not level, is the clearer path for future features.

Honest limitation: the top drivers describe where a county is and what it already was, not what is currently changing. The model excels at confirming known-vulnerable counties and is weaker at catching a new crisis in a previously stable one, a caveat we carry forward, not a reason to discount the result.

RECOMMENDATIONS

From notebook to NDMA operations

Deploy as a monthly early-warning bulletin

Scheduled job rebuilds features as new CHIRPS & WFP data land; each county gets a crisis probability + a plain-language SHAP explanation on an NDMA dashboard.

Keep the analyst in the loop

Flags inform; they never trigger aid directly. Log every prediction and score it against the IPC release 4–8 weeks later to catch drift before it costs a missed crisis.

Next feature work: dynamics over statics

Add price trends (not levels) and longer rolling rainfall windows to improve detection of new crises emerging in previously stable counties.

Scale the architecture

CHIRPS, WFP prices and FEWS NET IPC exist across East Africa, so the same pipeline can extend beyond Kenya.

The prediction problem is solved to the standard NDMA needs; what remains is operational integration.

Thank you.

From reactive response to predictive protection, weeks earlier, for the counties that need it most.

Questions & discussion